

Deliverable 1 Report

Deliverable 1 was around setting up the Python code for pre-processing reference data, creating the model, and fine-tuning the model. The model refers to the Moderate Spatial Resolution Burned Area Change Detection (MSR-BACD) model developed by Liu et al. (2025), which we will be training on our reference data to produce predictions of changes in Portugal. The initial model, along with processes for training it with either 6 or 10 spectral bands and increasing the number of output classes, has been uploaded to the scripts/ directory in the project GitHub.

For pre-processing reference data, we needed HDF5 files of all the Sentinel-2 tiles of Portugal. I created the scripts for both creating and appending additional data to these HDF5 files and tested the output data was correct. Through testing, I also identified changes to improve the processing efficiency and reduce the output file size and implemented these changes. These scripts are also in the scripts/ directory of the project GitHub, and were passed along to CACN so that they could create the HDF5 files for our project.

With the HDF5 files now created by CACN and the model code usable for our purposes, we can now move on with training the model and running it on full Sentinel-2 tiles.

Project GitHub link - https://github.com/S2change/vegetation_loss

Citations

Liu, P., Liu, Y., Liu, Y., Guo, X., Liu, Y., Liu, Y., Zhao, W., & Xu, W. (2025). Faster, better, and more accurate mapping of burned areas using Sentinel-2 multispectral images. *Remote Sensing of Environment*, 333, 115137. <https://doi.org/10.1016/j.rse.2025.115137>